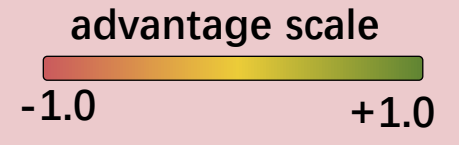


TreeMAPPO VS Existing Methods

Example Question: What is $3 \times 2 + 1$?

GRPO (No Branching): Coarse Credit Assignment

- Q — $3 \times 2 = 6$. $6 + 1 = \boxed{7}$ ✓ correctly assigned
— $3 \times 2 = 42$. $42 + 1 = \boxed{43}$ ⚠ should not blame the last step
— $3 \times 2 = 6$, $6 + 1 = \boxed{42}$ ⚠ should not blame the first step
— $3 \times 2 = 42$, wait, actually, $3 \times 2 = 6$, $6 + 1 = \boxed{7}$ ⚠ should not credit the first step



Spontaneous Branching: Trajectory diverges too soon

- Q — OK. Let's do it step by step... ⚠ Produces no meaningful signals
— us do it step by step...
— Let's first calculate...

Win-Rate or Heuristic-Based Credit Assignment: Bias credits towards leaves

- Q — $3 \times 2 = 6$ — $6 + 1 = \boxed{7}$ ✓ High advantage contrast for sibling correctness divergence
— $6 + 1 = \boxed{42}$
— $6 + 1 = 7$. Therefore, the answer is $\boxed{7}$. ⚠ Aggressively assign leaf advantages, but they may not be the core contributors.
— So, $3 \times 2 + 1 = \boxed{7}$.

Our TreeMAPPO Algorithm: Curated Branching Policy and Advantage Calculation

- Q — $3 \times 2 = 6$ — $6 + 1 = \boxed{7}$ ✓ High advantage contrast for sibling correctness divergence
— $6 + 1 = \boxed{42}$
— $6 + 1 = 7$. Therefore, the answer is $\boxed{7}$. ✓ If leaf siblings agree, the contributor is more likely parent than leaves
— So, $3 \times 2 + 1 = \boxed{7}$.